

Part I

SUMMARY of Derivatives of Inverse Functions

Summary

February 21, 2026

Summary

$$\begin{aligned}
 & \bullet \frac{d}{dx} [\sin^{-1}(x)] = \frac{1}{\sqrt{1-x^2}} & \bullet \frac{d}{dx} [\cos^{-1}(x)] = \frac{-1}{\sqrt{1-x^2}} \\
 & \bullet \frac{d}{dx} [\tan^{-1}(x)] = \frac{1}{1+x^2} & \bullet \frac{d}{dx} [\cot^{-1}(x)] = \frac{-1}{1+x^2} \\
 & \bullet \frac{d}{dx} [\sec^{-1}(x)] = \frac{1}{|x|\sqrt{x^2-1}} \text{ for } |x| \geq 1 & \bullet \frac{d}{dx} [\csc^{-1}(x)] = \frac{-1}{|x|\sqrt{x^2-1}} \text{ for } |x| \geq 1
 \end{aligned}$$

Theorem 1 (The Inverse Function Theorem). If f is a differentiable function that is one-to-one near a and $f'(a) \neq 0$, then:

- (a) $f^{-1}(x)$ is **defined** for x near $b = f(a)$,
- (b) $f^{-1}(x)$ is **differentiable** for x near $b = f(a)$,
- (c) last, but not least: $\left[\frac{d}{dx} f^{-1}(x) \right]_{x=b} = \frac{1}{f'(a)}$ where $b = f(a)$.

Part II

Recitation Questions

Problem 1

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Problem 1 Explain what each of the following means:

(a) $\sin^{-1}(x)$

Explanation. This denotes the inverse function to $\sin(x)$, sometimes denoted by $\arcsin(x)$.

(b) $(\sin(x))^{-1}$

Explanation. This means $\sin(x)$ raised to the -1 power, i.e. $\frac{1}{\sin(x)}$.

(c) $\sin(x^{-1})$

Explanation. This means $\sin\left(\frac{1}{x}\right)$.

Problem 2

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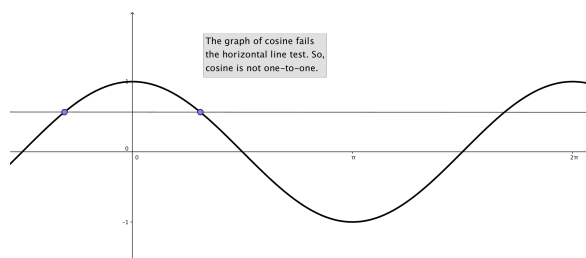
Problem 2

Problem 2 Without using a calculator, determine if the statement below is true or false.

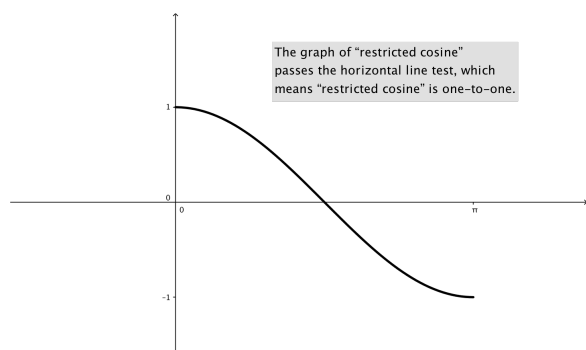
$$\cos^{-1}(\cos(7\pi/6)) = 7\pi/6$$

Explanation. This statement is **false**: the correct statement is $\cos^{-1}(\cos(7\pi/6)) = 5\pi/6$. (Why?)

Spoiler Alert: the cosine function is *not* invertible since its graph fails the horizontal line test.



To produce the inverse cosine we must first restrict the domain of cosine, to the interval $[0, \pi]$, to produce an invertible function:



So, the range of $\cos^{-1}$ is $[0, \pi]$. Since $7\pi/6$ is not in this range, $7\pi/6$ is *never* a possible output of $\cos^{-1}$.

Problem 3

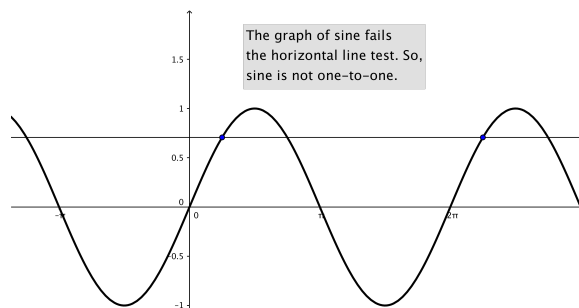
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Problem 3

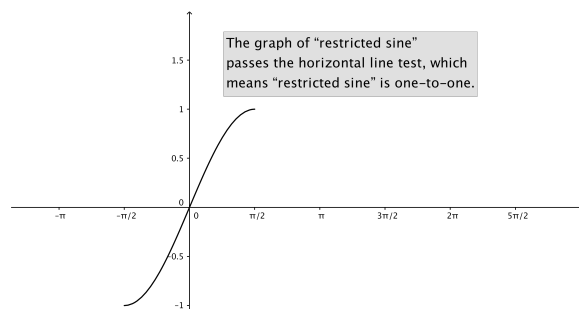
Problem 3 True or False: $\sin^{-1}(0) = \pi$.

Explanation. This statement is **false**: the correct statement is $\sin^{-1}(0) = 0$. (Why?)

Spoiler Alert: the sine function is *not* invertible since its graph fails the horizontal line test.



To produce the inverse sine we first restrict the domain of sine, to the interval $[-\pi/2, \pi/2]$, to produce an invertible function:



So, the range of $\sin^{-1}$ is $[-\pi/2, \pi/2]$. Since π is not in this range, π , is *never* a possible output of $\sin^{-1}$.

Problem 4

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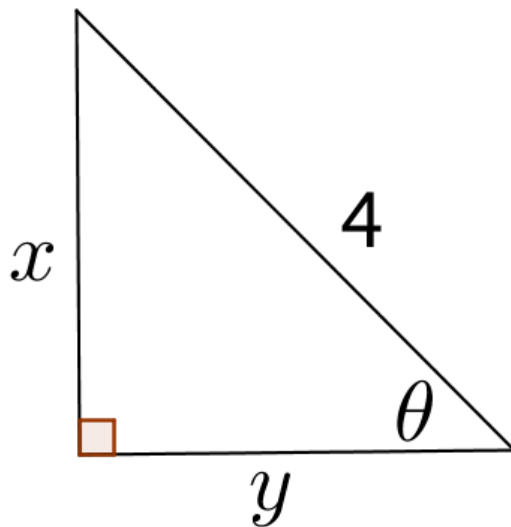
Problem 4 Simplify each of the following expressions.

(a) $\cos^{-1}(\sin(\pi/2))$

Explanation. By the unit circle, $\sin(\pi/2) = 1$, and so we are looking for $\cos^{-1}(1)$. The range of $\cos^{-1}$ is $[0, \pi]$, and so, by properties of inverse functions, $\cos^{-1}(1) = 0$.

(b) $\tan(\sin^{-1}(x/4))$

Explanation. Let $\theta = \sin^{-1}(x/4)$, then $\sin(\theta) = x/4$. We can then draw the corresponding right triangle:



Calling the adjacent side y , by the Pythagorean Theorem we obtain

$$4^2 = x^2 + y^2 \implies y = \sqrt{16 - x^2}$$

Remark: Since θ is in the range of $\sin^{-1}$, it follows that $-\pi/2 < \theta < \pi/2$. Therefore, $\cos(\theta) = y/4 > 0$. Therefore, $y > 0$.
Then

$$\begin{aligned} \tan(\sin^{-1}(x/4)) &= \tan(\theta), \\ &= \frac{x}{y}, \\ &= \frac{x}{\sqrt{16 - x^2}}. \end{aligned}$$

Note: $\tan \theta$ has the same sign as x , since $y > 0$.

Problem 5

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Problem 5 A table of values for f and f' is shown below. Suppose that f is a one-to-one function and f^{-1} is its inverse.

x	$f(x)$	$f'(x)$
1	3	4
3	4	5
4	6	3

- (a) Evaluate $f^{-1}(f(x))$ at $x = 3$.

Explanation. $f^{-1}(f(3)) = f^{-1}(4) = 3$

- (b) Evaluate $\frac{d}{dx}f(f(x))$ at $x = 3$.

Explanation.

$$\begin{aligned}\frac{d}{dx}f(f(x)) &= f'(f(x)) \cdot f'(x) \implies \left[\frac{d}{dx}f(f(x)) \right]_{x=3} = f'(f(3)) \cdot f'(3) \\ &= f'(4) \cdot 5 = 3 \cdot 5 = 15\end{aligned}$$

- (c) Evaluate $\frac{d}{dx}\ln(f(x))$ at $x = 3$. [2]

Explanation.

$$\begin{aligned}\frac{d}{dx}\ln(f(x)) &= \frac{f'(x)}{f(x)} \\ \implies \left[\frac{d}{dx}\ln(f(x)) \right]_{x=3} &= \frac{f'(3)}{f(3)} = \frac{5}{4}\end{aligned}$$

- (d) Evaluate $f^{-1}(x)$ at $x = 3$. [2]

Explanation. $f^{-1}(3) = 1 \iff 3 = f(1)$

- (e) Evaluate $\frac{d}{dx}f^{-1}(x)$ at $x = 3$. [2]

Explanation.

$$\begin{aligned}\frac{d}{dx}f^{-1}(x) &= \frac{1}{f'(f^{-1}(x))} \\ \implies \left[\frac{d}{dx}f^{-1}(x) \right]_{x=3} &= \frac{1}{f'(f^{-1}(3))} \\ &= \frac{1}{f'(1)} = \frac{1}{4}\end{aligned}$$

- (f) Evaluate $\lim_{x \rightarrow 4} \frac{f(x) - f(4)}{x - 4}$ [2]

Explanation. $\lim_{x \rightarrow 4} \frac{f(x) - f(4)}{x - 4} = f'(4) = 3$

Problem 6

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Problem 6

Problem 6 *An object is moving along a horizontal line. Its position (in meters) at the time t (in seconds) is given by $s(t)$. What does the value $s^{-1}(5)$ represent?*

Explanation. $s^{-1}(5)$ represents the the time when the position of the object is 5m.

Problem 7

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Problem 7 Given the expression for $f(x)$, find the derivative of f^{-1} at the given point on the graph of f^{-1} , without solving for f^{-1} .

- (a) $f(x) = x^2 + 1$ (for $x \geq 0$); the point on the graph of f^{-1} : $(5, 2)$.

Verify your answer by evaluating the derivative of f^{-1} at the given point.

Explanation. $(f^{-1})'(5) = \frac{1}{f'(2)}$. Since $f'(x) = 2x$, $f'(2) = 4$. Thus,

$$(f^{-1})'(5) = \frac{1}{4}.$$

$$\text{Verifying: } f^{-1}(x) = \sqrt{x-1} \implies \frac{d}{dx} f^{-1}(x) = \frac{1}{2\sqrt{x-1}} \implies \left[\frac{d}{dx} f^{-1} \right]_{x=5} = \frac{1}{2\sqrt{5-1}} = \frac{1}{4}$$

- (b) $f(x) = x^2 - 2x - 3$ (for $x \leq 1$); the point on the graph of f^{-1} : $(12, -3)$.

Explanation. $(f^{-1})'(12) = \frac{1}{f'(-3)}$. Since $f'(x) = 2x - 2$, $f'(-3) = -6 - 2 = -8$. Thus, $(f^{-1})'(12) = -\frac{1}{8}$.

Notice that for this problem, finding the inverse of $f(x)$ involves complicated algebra. In this case, it's much easier to find the derivative of f^{-1} without solving for f^{-1} .

Problem 8

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Problem 8

Problem 8 Find the slope of the tangent line to the curve $y = f^{-1}(x)$ at $(4, 7)$ if the slope of the tangent line to the curve $y = f(x)$ at $(7, 4)$ is $\frac{2}{3}$.

Explanation. Note that the statement “the slope of the tangent line to the curve $y = f(x)$ at $(7, 4)$ is $\frac{2}{3}$ ” specifically means that $f'(7) = \frac{2}{3}$. The slope of the tangent line to the curve $y = f^{-1}(x)$ at $(4, 7)$ is $(f^{-1})'(4)$, and so we use the formula for the derivative of the inverse function to compute: $(f^{-1})'(4) = \frac{1}{f'(7)} = \frac{1}{\frac{2}{3}} = \frac{3}{2}$.

Problem 9

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Problem 9 Find the derivatives of the following functions:

(a) $f(x) = \sec^{-1}(\sqrt{x})$.

Explanation. $f'(x) = \frac{1}{\sqrt{x}\sqrt{x-1}} \cdot \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2x\sqrt{x-1}}$

(b) $g(x) = \ln(\sin^{-1}(x))$.

Explanation. $g'(x) = \frac{1}{\sin^{-1}(x)} \cdot \frac{1}{\sqrt{1-x^2}}$.

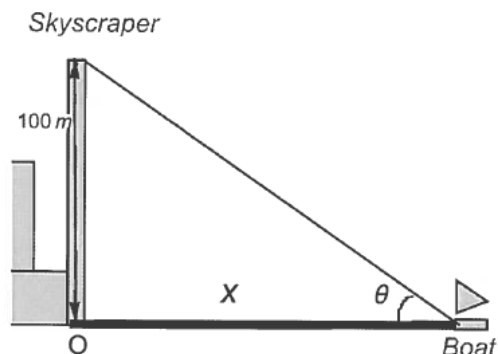
(c) $h(x) = \frac{1}{\tan^{-1}(x^2 + 4)}$.

Explanation. $h'(x) = -(\tan^{-1}(x^2 + 4))^{-2} \cdot \frac{1}{1 + (x^2 + 4)^2} \cdot (2x)$.

Problem 10

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Problem 10 A boat sails directly toward a 100-meter skyscraper that stands on the edge of a harbor. The angular size θ of the building is the angle formed by lines from the top and bottom of the building to the observer on the boat (see figure below).



- (a) Express the angle θ as the function of x , the distance of the boat from the building.

Explanation. $\tan \theta = \frac{100}{x} \implies \theta = \tan^{-1} \left(\frac{100}{x} \right)$

- (b) The boat is sailing directly toward the skyscraper at 3 m/s. Find $\frac{d\theta}{dt}$ when the boat is $x = 300\text{m}$ from the building. [3]

Explanation.

$$\begin{aligned} \frac{d\theta}{dt} &= \frac{d}{dt} \tan^{-1} \left(\frac{100}{x} \right) \\ &= \frac{d}{dx} \tan^{-1} \left(\frac{100}{x} \right) \frac{dx}{dt} \\ &= \frac{1}{1 + \left(\frac{100}{x} \right)^2} \cdot \frac{-100}{x^2} \frac{dx}{dt} \\ &= \frac{1}{1 + \left(\frac{100}{x} \right)^2} \cdot \frac{-100}{x^2} \cdot (-3) \\ &= \frac{300}{x^2 + 100^2} \end{aligned}$$

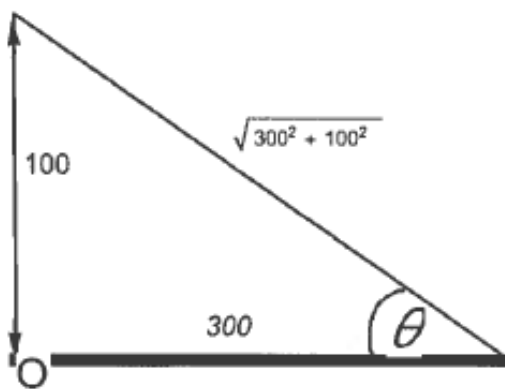
Problem 10

$$\begin{aligned}
 \left[\frac{d\theta}{dt} \right]_{x=300} &= \frac{300}{300^2 + 100^2} \\
 &= \frac{3}{900 + 100} \\
 &= \frac{3}{1000} \\
 &= 0.003 \text{ rad/sec}
 \end{aligned}$$

Alternatively, this problem could be done without inverse functions.

$$\begin{aligned}
 \tan \theta &= \frac{100}{x} \\
 \frac{d}{dt}(\tan \theta) &= \frac{d}{dt} \left(\frac{100}{x} \right) \\
 \sec^2 \theta \frac{d\theta}{dt} &= \frac{-100}{x^2} \frac{dx}{dt}
 \end{aligned}$$

We have to evaluate $\sec^2 \theta$ at the moment when $x = 300\text{m}$



Using the triangle above, we have:

$$\sec \theta = \frac{\sqrt{300^2 + 100^2}}{300} = 100 \frac{\sqrt{3^2 + 1^2}}{300} = \frac{\sqrt{10}}{3}, \text{ we obtain:}$$

$$\begin{aligned}
 \left(\frac{\sqrt{10}}{3} \right)^2 \left[\frac{d\theta}{dt} \right]_{x=300} &= -\frac{100}{300^2} (-3) \\
 \left[\frac{d\theta}{dt} \right]_{x=300} &= \frac{1}{300} \cdot \frac{9}{10} = \frac{3}{1000} \text{ rad/sec}
 \end{aligned}$$